DATABASE TECHNOLOGIES – ASSIGNMENT 7 OPTIONAL

1.Display department name and manager first name.

SELECT d.DEPARTMENT_NAME, e.FIRST_NAME

FROM departments d

INNER JOIN employees e

ON d.MANAGER_ID = e.EMPLOYEE_ID;

```
[d5_96525_Deven@hr]>SELECT d.DEPARTMENT_NAME, e.FIRST_NAME
    -> FROM departments d
    -> INNER JOIN employees e
    -> ON d.MANAGER_ID = e.EMPLOYEE_ID;
+------------------+------------+
| DEPARTMENT_NAME  | FIRST_NAME |
+------------------+------------+
| Administration   | Jennifer   |
| Marketing        | Michael    |
| Purchasing       | Den        |
| Human Resources  | Susan      |
| Shipping         | Adam       |
| IT               | Alexander  |
| Public Relations | Hermann    |
| Sales            | John       |
| Executive        | Steven     |
| Finance          | Nancy      |
| Accounting       | Shelley    |
+------------------+------------+
11 rows in set (0.008 sec)
```

2.Display department name, manager name, and city

SELECT d.DEPARTMENT_NAME, e.FIRST_NAME,l.city

FROM departments d

INNER JOIN employees e

ON d.MANAGER_ID = e.EMPLOYEE_ID

INNER JOIN locations l

ON d.LOCATION_ID=l.LOCATION_ID;

```
[d5_96525_Deven@hr]>SELECT d.DEPARTMENT_NAME, e.FIRST_NAME,l.city
    -> FROM departments d
    -> INNER JOIN employees e
    -> ON d.MANAGER_ID = e.EMPLOYEE_ID
    -> INNER JOIN locations l
    -> ON d.LOCATION_ID=l.LOCATION_ID;
+------------------+------------+---------------------+
| DEPARTMENT_NAME  | FIRST_NAME | city                |
+------------------+------------+---------------------+
| Administration   | Jennifer   | Seattle             |
| Marketing        | Michael    | Toronto             |
| Purchasing       | Den        | Seattle             |
| Human Resources  | Susan      | London              |
| Shipping         | Adam       | South San Francisco |
| IT               | Alexander  | Southlake           |
| Public Relations | Hermann    | Munich              |
| Sales            | John       | Oxford              |
| Executive        | Steven     | Seattle             |
| Finance          | Nancy      | Seattle             |
| Accounting       | Shelley    | Seattle             |
+------------------+------------+---------------------+
11 rows in set (0.008 sec)
```

3.Display country name, city, and department name.

SELECT c.COUNTRY_NAME,l.CITY,d.DEPARTMENT_NAME

FROM locations l INNER JOIN departments d

ON d.LOCATION_ID=l.LOCATION_ID

INNER JOIN countries c

ON l.COUNTRY_ID=c.COUNTRY_ID;

```
[d5_96525_Deven@hr]>SELECT c.COUNTRY_NAME,l.CITY,d.DEPARTMENT_NAME
    -> FROM locations l INNER JOIN departments d
    -> ON d.LOCATION_ID=l.LOCATION_ID
    -> INNER JOIN countries c
    -> ON l.COUNTRY_ID=c.COUNTRY_ID;
+----------------------------------------+---------------------+----------------------+
| COUNTRY_NAME                           | CITY                | DEPARTMENT_NAME      |
+----------------------------------------+---------------------+----------------------+
| United States of America               | Seattle             | Administration       |
| Canada                                 | Toronto             | Marketing            |
| United States of America               | Seattle             | Purchasing           |
| United Kingdom of Great Britain and Nort | London              | Human Resources      |
| United States of America               | South San Francisco | Shipping             |
| United States of America               | Southlake           | IT                   |
| Germany                                | Munich              | Public Relations     |
| United Kingdom of Great Britain and Nort | Oxford              | Sales                |
| United States of America               | Seattle             | Executive            |
| United States of America               | Seattle             | Finance              |
| United States of America               | Seattle             | Accounting           |
| United States of America               | Seattle             | Treasury             |
| United States of America               | Seattle             | Corporate Tax        |
| United States of America               | Seattle             | Control And Credit   |
| United States of America               | Seattle             | Shareholder Services |
| United States of America               | Seattle             | Benefits             |
| United States of America               | Seattle             | Manufacturing        |
| United States of America               | Seattle             | Construction         |
| United States of America               | Seattle             | Contracting          |
| United States of America               | Seattle             | Operations           |
| United States of America               | Seattle             | IT Support           |
| United States of America               | Seattle             | NOC                  |
| United States of America               | Seattle             | IT Helpdesk          |
| United States of America               | Seattle             | Government Sales     |
| United States of America               | Seattle             | Retail Sales         |
| United States of America               | Seattle             | Recruiting           |
| United States of America               | Seattle             | Payroll              |
+----------------------------------------+---------------------+----------------------+
27 rows in set (0.118 sec)
```

4.Display job title, department name, employee last name, starting date for all jobs

from 2011 to 2015.

```
SELECT
   j.JOB_TITLE ,
   d.DEPARTMENT_NAME,
   e.LAST_NAME,
   jh.START_DATE
FROM employees e
INNER JOIN jobs j ON e.JOB_ID=j.JOB_ID
INNER JOIN job_history jh ON e.EMPLOYEE_ID = jh.employee_id
INNER JOIN departments d ON d.DEPARTMENT_ID = e.department_id
WHERE YEAR(jh.START_DATE) >= 2011 AND YEAR(jh.START_DATE) <= 2015;
```

```
[d5_96525_Deven@hr]>SELECT
    ->     j.JOB_TITLE ,
    ->     d.DEPARTMENT_NAME,
    ->     e.LAST_NAME,
    ->     jh.START_DATE
    -> FROM employees e
    -> INNER JOIN jobs j ON e.JOB_ID=j.JOB_ID
    -> INNER JOIN job_history jh ON e.EMPLOYEE_ID = jh.employee_id
    -> INNER JOIN departments d ON d.DEPARTMENT_ID = e.department_id
    -> WHERE YEAR(jh.START_DATE) >= 2011 AND YEAR(jh.START_DATE) <= 2015;
+-------------------------------+-----------------+-----------+------------+
| JOB_TITLE                     | DEPARTMENT_NAME | LAST_NAME | START_DATE |
+-------------------------------+-----------------+-----------+------------+
| Marketing Manager             | Marketing       | Martinez  | 2014-02-17 |
| Administration Vice President | Executive       | Garcia    | 2011-01-13 |
| Administration Assistant      | Administration  | Whalen    | 2012-07-01 |
| Administration Vice President | Executive       | Yang      | 2011-10-28 |
+-------------------------------+-----------------+-----------+------------+
4 rows in set (0.011 sec)
```

5.Display job title and average salary of employees.

SELECT

j.JOB_TITLE ,

AVG(e.salary)

FROM employees e

INNER JOIN jobs j ON e.JOB_ID=j.JOB_ID

GROUP BY j.JOB_TITLE;

```
[d5_96525_Deven@hr]>SELECT
    ->  j.JOB_TITLE ,
    ->  AVG(e.salary)
    -> FROM employees e
    -> INNER JOIN jobs j ON e.JOB_ID=j.JOB_ID
    -> GROUP BY j.JOB_TITLE;
+---------------------------------+---------------+
| JOB_TITLE                       | AVG(e.salary) |
+---------------------------------+---------------+
| Public Accountant               |   8300.000000 |
| Accounting Manager              |  12008.000000 |
| Administration Assistant        |   4400.000000 |
| President                       |  24000.000000 |
| Administration Vice President   |  17000.000000 |
| Accountant                      |   7920.000000 |
| Finance Manager                 |  12008.000000 |
| Human Resources Representative  |   6500.000000 |
| Programmer                      |   5760.000000 |
| Marketing Manager               |  13000.000000 |
| Marketing Representative        |   6000.000000 |
| Public Relations Representative |  10000.000000 |
| Purchasing Clerk                |   2780.000000 |
| Purchasing Manager              |  11000.000000 |
| Sales Manager                   |  12200.000000 |
| Sales Representative            |   8350.000000 |
| Shipping Clerk                  |   3215.000000 |
| Stock Clerk                     |   2785.000000 |
| Stock Manager                   |   7280.000000 |
+---------------------------------+---------------+
19 rows in set (0.147 sec)
```

6.Display job title, employee name, and the difference between maximum salary

for that job and salary of the employee.

```
[d5_96525_Deven@hr]>SELECT j.JOB_TITLE, e.FIRST_NAME,(j.MAX_SALARY - e.salary) FROM employees e
    -> INNER JOIN jobs j ON e.JOB_ID=j.JOB_ID;
+---------------------------------+-------------+--------------------------+
| JOB_TITLE                       | FIRST_NAME  | (j.MAX_SALARY - e.salary) |
+---------------------------------+-------------+--------------------------+
| Public Accountant               | William     |                   700.00 |
| Accounting Manager              | Shelley     |                  3992.00 |
| Administration Assistant        | Jennifer    |                  1600.00 |
| President                       | Steven      |                 16000.00 |
| Administration Vice President   | Neena       |                 13000.00 |
| Administration Vice President   | Lex         |                 13000.00 |
| Accountant                      | Daniel      |                     0.00 |
| Accountant                      | John        |                   800.00 |
| Accountant                      | Ismael      |                  1300.00 |
| Accountant                      | Jose Manuel |                  1200.00 |
| Accountant                      | Luis        |                  2100.00 |
| Finance Manager                 | Nancy       |                  3992.00 |
| Human Resources Representative  | Susan       |                  2500.00 |
| Programmer                      | Alexander   |                  1000.00 |
| Programmer                      | Bruce       |                  4000.00 |
| Programmer                      | David       |                  5200.00 |
| Programmer                      | Valli       |                  5200.00 |
| Programmer                      | Diana       |                  5800.00 |
| Marketing Manager               | Michael     |                  2000.00 |
| Marketing Representative        | Pat         |                  3000.00 |
| Public Relations Representative | Hermann     |                   500.00 |
| Purchasing Clerk                | Alexander   |                  2400.00 |
| Purchasing Clerk                | Shelli      |                  2600.00 |
| Purchasing Clerk                | Sigal       |                  2700.00 |
| Purchasing Clerk                | Guy         |                  2900.00 |
| Purchasing Clerk                | Karen       |                  3000.00 |
| Purchasing Manager              | Den         |                  4000.00 |
| Sales Manager                   | John        |                  6080.00 |
| Sales Manager                   | Karen       |                  6580.00 |
```

7.Display the last name and job title of employees who earn a commission

but do not belong to any department.

SELECT j.JOB_TITLE, e.LAST_NAME

FROM employees e

INNER JOIN jobs j ON e.JOB_ID = j.JOB_ID

WHERE e.COMMISSION_PCT IS NOT NULL AND e.DEPARTMENT_ID IS NULL;

```
[d5_96525_Deven@hr]>SELECT j.JOB_TITLE, e.LAST_NAME
    -> FROM employees e
    -> INNER JOIN jobs j ON e.JOB_ID = j.JOB_ID
    -> WHERE e.COMMISSION_PCT IS NOT NULL
    ->   AND e.DEPARTMENT_ID IS NULL;
+----------------------+-----------+
| JOB_TITLE            | LAST_NAME |
+----------------------+-----------+
| Sales Representative | Grant     |
+----------------------+-----------+
1 row in set (0.144 sec)
```

8.Display details of jobs that were done by any employee who is currently

drawing more than 15000 of salary.

SELECT j.job_id,

j.job_title,

e.last_name,

e.salary

FROM employees e

INNER JOIN jobs j ON e.JOB_ID = j.JOB_ID

WHERE e.salary>15000;

```
[d5_96525_Deven@hr]>SELECT  j.job_id,
    ->  j.job_title,
    ->  e.last_name,
    ->  e.salary
    -> FROM employees e
    -> INNER JOIN jobs j ON e.JOB_ID = j.JOB_ID
    -> WHERE e.salary>15000;
+---------+------------------------------+-----------+----------+
| job_id  | job_title                    | last_name | salary   |
+---------+------------------------------+-----------+----------+
| AD_PRES | President                    | King      | 24000.00 |
| AD_VP   | Administration Vice President | Yang      | 17000.00 |
| AD_VP   | Administration Vice President | Garcia    | 17000.00 |
+---------+------------------------------+-----------+----------+
3 rows in set (0.126 sec)
```

9.Display department name, manager name, and salary of the

managers, who have the experience of more than 12 years.

```
SELECT
   d.DEPARTMENT_NAME,
   e.LAST_NAME,
   e.salary
FROM departments d
INNER JOIN employees e ON d.MANAGER_ID = e.EMPLOYEE_ID
WHERE TIMESTAMPDIFF(YEAR, e.HIRE_DATE, NOW()) > 12;
```

```
[d5_96525_Deven@hr]>SELECT      d.DEPARTMENT_NAME,
    ->  e.LAST_NAME,
    ->  e.salary
    -> FROM departments d
    -> INNER JOIN employees e
    -> ON d.MANAGER_ID = e.EMPLOYEE_ID
    -> WHERE TIMESTAMPDIFF(YEAR, e.HIRE_DATE, NOW()) > 12;
+------------------+-----------+----------+
| DEPARTMENT_NAME  | LAST_NAME | salary   |
+------------------+-----------+----------+
| Purchasing       | Li        | 11000.00 |
| Human Resources  | Jacobs    |  6500.00 |
| Public Relations | Brown     | 10000.00 |
| Finance          | Gruenberg | 12008.00 |
| Accounting       | Higgins   | 12008.00 |
+------------------+-----------+----------+
5 rows in set (0.121 sec)
```

10. Display employee name if the employee joined before his manager.

```
SELECT  t1.LAST_NAME AS EMP,t1.HIRE_DATE,t2.LAST_NAME AS MGR,T2.HIRE_DATE
FROM employees t1
INNER JOIN employees t2
ON t1.MANAGER_ID = t2.EMPLOYEE_ID
WHERE t1.HIRE_DATE<t2.HIRE_DATE;
```

```
[d5_96525_Deven@hr]>SELECT  t1.LAST_NAME AS EMP,t1.HIRE_DATE,t2.LAST_NAME AS MGR,T2.HIRE_DATE
    -> FROM employees t1
    -> INNER JOIN employees t2
    -> ON t1.MANAGER_ID = t2.EMPLOYEE_ID
    -> WHERE t1.HIRE_DATE<t2.HIRE_DATE;
+------------+------------+-----------+------------+
| EMP        | HIRE_DATE  | MGR       | HIRE_DATE  |
+------------+------------+-----------+------------+
| Garcia     | 2011-01-13 | King      | 2013-06-17 |
| Williams   | 2015-06-25 | James     | 2016-01-03 |
| Gruenberg  | 2012-08-17 | Yang      | 2015-09-21 |
| Faviet     | 2012-08-16 | Gruenberg | 2012-08-17 |
| Li         | 2012-12-07 | King      | 2013-06-17 |
| Kaufling   | 2013-05-01 | King      | 2013-06-17 |
| Marlow     | 2015-02-16 | Fripp     | 2015-04-10 |
| Ladwig     | 2013-07-14 | Vollman   | 2015-10-10 |
| Rajs       | 2013-10-17 | Mourgos   | 2017-11-16 |
| Davies     | 2015-01-29 | Mourgos   | 2017-11-16 |
| Matos      | 2016-03-15 | Mourgos   | 2017-11-16 |
| Vargas     | 2016-07-09 | Mourgos   | 2017-11-16 |
| King       | 2014-01-30 | Partners  | 2015-01-05 |
| Sully      | 2014-03-04 | Partners  | 2015-01-05 |
| McEwen     | 2014-08-01 | Partners  | 2015-01-05 |
| Ozer       | 2015-03-11 | Cambrault | 2017-10-15 |
| Bloom      | 2016-03-23 | Cambrault | 2017-10-15 |
| Fox        | 2016-01-24 | Cambrault | 2017-10-15 |
| Smith      | 2017-02-23 | Cambrault | 2017-10-15 |
| Bates      | 2017-03-24 | Cambrault | 2017-10-15 |
| Abel       | 2014-05-11 | Zlotkey   | 2018-01-29 |
| Hutton     | 2015-03-19 | Zlotkey   | 2018-01-29 |
| Taylor     | 2016-03-24 | Zlotkey   | 2018-01-29 |
| Livingston | 2016-04-23 | Zlotkey   | 2018-01-29 |
| Grant      | 2017-05-24 | Zlotkey   | 2018-01-29 |
```

11.Display the employee’s name, job title for the jobs, employee did in the past where the job was done less than 12 months.

```
SELECT  e.LAST_NAME,
        j.JOB_TITLE,
        TIMESTAMPDIFF(MONTH, jh.START_DATE, jh.END_DATE)
FROM employees e
INNER JOIN job_history jh
ON e.EMPLOYEE_ID = jh.EMPLOYEE_ID
INNER JOIN jobs j
ON jh.JOB_ID = j.JOB_ID
WHERE TIMESTAMPDIFF(MONTH, jh.START_DATE, jh.END_DATE) < 12;
```

```
[d5_96525_Deven@hr]>SELECT  e.LAST_NAME,
    ->  j.JOB_TITLE,
    ->  TIMESTAMPDIFF(MONTH, jh.START_DATE, jh.END_DATE)
    -> FROM employees e
    -> INNER JOIN job_history jh
    -> ON e.EMPLOYEE_ID = jh.EMPLOYEE_ID
    -> INNER JOIN jobs j
    -> ON jh.JOB_ID = j.JOB_ID
    -> WHERE TIMESTAMPDIFF(MONTH, jh.START_DATE, jh.END_DATE) < 12;
+-----------+----------------------+-------------------------------------------------+
| LAST_NAME | JOB_TITLE            | TIMESTAMPDIFF(MONTH, jh.START_DATE, jh.END_DATE) |
+-----------+----------------------+-------------------------------------------------+
| Kaufling  | Stock Clerk          |                                              11 |
| Taylor    | Sales Representative |                                               9 |
| Taylor    | Sales Manager        |                                              11 |
+-----------+----------------------+-------------------------------------------------+
3 rows in set (0.014 sec)
```

12. Display employee name and country in which he is working.

SELECT e.LAST_NAME,

c.COUNTRY_NAME

FROM employees e

INNER JOIN departments d

ON e.DEPARTMENT_ID=d.DEPARTMENT_ID

INNER JOIN locations l

ON d.LOCATION_ID=l.LOCATION_ID

INNER JOIN countries c

ON l.COUNTRY_ID=c.COUNTRY_ID;

```
[d5_96525_Deven@hr]>SELECT  e.LAST_NAME,
    ->  c.COUNTRY_NAME
    -> FROM employees e
    -> INNER JOIN departments d
    -> ON e.DEPARTMENT_ID=d.DEPARTMENT_ID
    -> INNER JOIN locations l
    -> ON d.LOCATION_ID=l.LOCATION_ID
    -> INNER JOIN countries c
    -> ON l.COUNTRY_ID=c.COUNTRY_ID;
+-------------+-----------------------------------------+
| LAST_NAME   | COUNTRY_NAME                            |
+-------------+-----------------------------------------+
| James       | United States of America                |
| Miller      | United States of America                |
| Williams    | United States of America                |
| Jackson     | United States of America                |
| Nguyen      | United States of America                |
| Weiss       | United States of America                |
| Fripp       | United States of America                |
| Kaufling    | United States of America                |
| Vollman     | United States of America                |
| Mourgos     | United States of America                |
| Nayer       | United States of America                |
| Mikkilineni | United States of America                |
| Landry      | United States of America                |
| Markle      | United States of America                |
| Bissot      | United States of America                |
| Atkinson    | United States of America                |
| Marlow      | United States of America                |
| Olson       | United States of America                |
| Mallin      | United States of America                |
```

13.Display department name, average salary and number of employees

with commission within the department.

```
SELECT  d.DEPARTMENT_NAME,
        AVG(e.salary) AS AVG_SAL,
        COUNT(e.EMPLOYEE_ID) AS EMP_COUNT
FROM employees e INNER JOIN departments d
ON e.DEPARTMENT_ID=d.DEPARTMENT_ID
GROUP BY d.DEPARTMENT_NAME;
```

```
[d5_96525_Deven@hr]>SELECT  d.DEPARTMENT_NAME,
    ->  AVG(e.salary) AS AVG_SAL,
    ->  COUNT(e.EMPLOYEE_ID) AS EMP_COUNT
    -> FROM employees e INNER JOIN departments d
    -> ON e.DEPARTMENT_ID=d.DEPARTMENT_ID
    -> GROUP BY d.DEPARTMENT_NAME;
+------------------+--------------+-----------+
| DEPARTMENT_NAME  | AVG_SAL      | EMP_COUNT |
+------------------+--------------+-----------+
| Executive        | 19333.333333 |         3 |
| IT               |  5760.000000 |         5 |
| Finance          |  8601.333333 |         6 |
| Purchasing       |  4150.000000 |         6 |
| Shipping         |  3475.555556 |        45 |
| Sales            |  8955.882353 |        34 |
| Administration   |  4400.000000 |         1 |
| Marketing        |  9500.000000 |         2 |
| Human Resources  |  6500.000000 |         1 |
| Public Relations | 10000.000000 |         1 |
| Accounting       | 10154.000000 |         2 |
+------------------+--------------+-----------+
11 rows in set (0.013 sec)
```

14.Display the count of employees, month and the year in which 3 or more employees joined in any department located in ‘Seattle’.

```
SELECT
   COUNT(e.EMPLOYEE_ID) ,
   MONTH(e.HIRE_DATE) ,
   YEAR(e.HIRE_DATE)
FROM employees e
INNER JOIN departments d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
INNER JOIN locations l ON d.LOCATION_ID = l.LOCATION_ID
WHERE l.CITY = 'Seattle'
GROUP BY YEAR(e.HIRE_DATE), MONTH(e.HIRE_DATE)
```

HAVING COUNT(e.EMPLOYEE_ID) >= 3;

```
[d5_96525_Deven@hr]>SELECT
    ->     COUNT(e.EMPLOYEE_ID) ,
    ->     MONTH(e.HIRE_DATE) ,
    ->     YEAR(e.HIRE_DATE)
    -> FROM employees e
    -> INNER JOIN departments d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
    -> INNER JOIN locations l ON d.LOCATION_ID = l.LOCATION_ID
    -> WHERE l.CITY = 'Seattle'
    -> GROUP BY YEAR(e.HIRE_DATE), MONTH(e.HIRE_DATE)
    -> HAVING COUNT(e.EMPLOYEE_ID) >= 3;
+----------------------+--------------------+-------------------+
| COUNT(e.EMPLOYEE_ID) | MONTH(e.HIRE_DATE) | YEAR(e.HIRE_DATE) |
+----------------------+--------------------+-------------------+
|                    3 |                  9 |              2015 |
+----------------------+--------------------+-------------------+
1 row in set (0.008 sec)
```

15.Display employee name, job title, start date, and end date of past jobs

of all employees with commission percentage null.

SELECT e.LAST_NAME,

j.JOB_TITLE,

jh.START_DATE,

jh.END_DATE

FROM employees e INNER JOIN jobs j

ON e.JOB_ID = j.JOB_ID

INNER JOIN job_history jh

ON e.EMPLOYEE_ID=jh.EMPLOYEE_ID

WHERE e.COMMISSION_PCT IS NULL;

```
[d5_96525_Deven@hr]>SELECT  e.LAST_NAME,
    ->  j.JOB_TITLE,
    ->  jh.START_DATE,
    ->  jh.END_DATE
    -> FROM employees e INNER JOIN jobs j
    -> ON e.JOB_ID = j.JOB_ID
    -> INNER JOIN job_history jh
    -> ON e.EMPLOYEE_ID=jh.EMPLOYEE_ID
    -> WHERE e.COMMISSION_PCT IS NULL;
+-----------+------------------------------+------------+------------+
| LAST_NAME | JOB_TITLE                    | START_DATE | END_DATE   |
+-----------+------------------------------+------------+------------+
| Yang      | Administration Vice President | 2007-09-21 | 2011-10-27 |
| Yang      | Administration Vice President | 2011-10-28 | 2015-03-15 |
| Garcia    | Administration Vice President | 2011-01-13 | 2016-07-24 |
| Li        | Purchasing Manager           | 2016-03-24 | 2017-12-31 |
| Kaufling  | Stock Manager                | 2017-01-01 | 2017-12-31 |
| Whalen    | Administration Assistant     | 2005-09-17 | 2011-06-17 |
| Whalen    | Administration Assistant     | 2012-07-01 | 2016-12-31 |
| Martinez  | Marketing Manager            | 2014-02-17 | 2017-12-19 |
+-----------+------------------------------+------------+------------+
8 rows in set (0.119 sec)

[d5_96525_Deven@hr]>
```